A Deep Learning Framework for Classifying Autism Spectrum Disorder from fMRI Images

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***Abstract*—The objective of this work is to propose a new machine learning based approach for ASD classification from fMRI data employing deep learning architecture and novel preprocessing method. Main datasets included in ABIDE I, ABIDE II, Human Connectome Project (HCP), and Social Functions and Autism Research Interface (SFARI). Many supplements include graph neural networks for brain segmentation, 3D-BRA for accurate mapping of the brain than just using conventional registration technique, EN–CC for feature selection. The approach incorporated graph neural network for segmentation of the brain, 3DBRA mapping that translates and rotates the brain precisely unlike the general technique of registration, EN-CC for selecting the features. The deep learning models under consideration were ResNet-50, VGG-16, Inception-v3, and Efficient Net which were improved through ABSC and MSTC. Classification accuracy of the proposed models was higher than 95% on all the datasets with Efficient Net scoring 99% on the SFARI dataset. This means that only the Stratified Group Cross-Validation (SGCV) was employed to make sure the results would be more reliable. The results present a considerable improvement in both accuracy and the interpretability of the derived connectivity profiles of the ASD-related brain.**

Keywords—Autism Spectrum Disorder, fMRI, Deep Learning, Feature Selection, Brain Connectivity

I. Introduction

Autism spectrum disorder (ASD) differential using Neuro imaging specifically functional magnetic resonance imaging (fMRI) has been widely considered over the recent years because of their promise of offering more reliable diagnostic information[1]. The ASD is a group of developmental disorders that are manifested in difficulties with social interaction, communication, and specific behaviours, and concerning which behavioural methods of diagnostics have been traditionally used. However, the huge variability of ASD is a major issue which complicates differentiation of this disease and its timely diagnose. Therefore, neuroimaging findings that reflects intricate patterns of connection appears as a valuable research aid for enhancing diagnosing and advancing ASD-associated brain mechanisms.

AI, specifically, deep learning has in the recent past demonstrated the ability in processing high dimensional neuroimaging data that include fMRI. These models can find out the small activation patterns of the brain connectivity which may possibly be having the characteristic of ASD and this has to be done without much focus on feature engineering. Despite the fact that there have been many studies on assessing ASD based on deep learning models, there are still some problems in the dimensionality and noise from fMRI data. Add to this, the standard models can fail to provide interpretability; thus, it is hard to comprehend what features of the neural networks lead to the outcomes achieved, a factor that is very crucial especially in the field of health. In this work, a new machine learning scheme is introduced for classifying ASD based on the current state of the art deep learning network models, preprocessing, and features extraction method tailored for neuroimaging datasets. In handling fMRI data, we overcome the complexity issue through the employment of GNNs for the segmentation of brain regions, besides feature selection with the novel EN-CC algorithm. Furthermore, by modifying such basic architectures as ResNet-50, Inception-v3 with the help of attention and multi-scale temporal convolution, we significantly boost the given models' capacity to capture dynamics of brain connectivity[2].

Our approach is to aim for high classification accuracy and simultaneously, model read-interpretability, which is applied to various datasets including ABIDE I, ABIDE II, HCP, and SFARI. To strengthen generalization attributes of the proposed models, we perform 3D Brain Region Alignment (3D-BRA) for the high level of model preprocessing and incorporate a new type of evaluation test known as Stratified Group Cross-Validation (SGCV). This research seeks to not only achieve classification improvement of ASD, but also to enhance diagnostic interpretation ability and find ASD related mechanisms in the brain[3]. Here, it is shown that preprocessing, feature selection advancements as well as employing deep learning algorithms in our work has resulted in an ASD classification accuracy which is well over 95% on multiple databases. Additionally, the employment of explainable AI methods guarantees that the produced models offer helpful information on the most involved brain areas in ASD for further neuroimaging research and potential clinical practice.

II. Literature Review

The diagnosis of Autism Spectrum Disorder (ASD) utilizing neuroimaging, especially functional Magnetic Resonance Imaging (fMRI) has changed over the last couple of years. Previous work mainly targeted identifying these connection patterns via machine learning techniques, most of which, however, encountered challenges in terms of processing the data with high dimension and nonlinearity. Therefore, early investigations employed methods of feature reduction including Principal Component Analysis (PCA) and Independent Component Analysis (ICA) to preprocess fMRI data, that were further classified by using classifiers: Support Vector Machines (SVMs) or Random Forests among others.

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Although these approaches gave rather moderate success, still these have some limitations, those included manual selection of features and inability to model complex dependencies in between different parts of the human brain[4].

Fortunately, with the advancement of deep learning, effective approaches to identifying automatically sufficiently important patterns from high-dimensional data became significantly better. Other types of deep learning models, called Convolutional Neural Networks (CNNs), which mainly served for image classification, were also introduced to neuroimaging, including ASD classification. Of these models, spatial patterns in the brain as they presented a way of enhancing the models also performed very well in terms of accuracy and robustness in comparison to traditional methods. Some of the CNNs' variants, including ResNet, VGG and Inception has been used in some of the studies and the results recorded an enhanced classification precision. These deep learning models have been further refined using methods; data augmentation, multimodal fusion, which is the integration of the fMRI data with other modalities such as EEG), as well as pre-training on large-scale Neuroimaging datasets[5].

Nonetheless, due to the high dimensionality of fMRI data new problems have persisted through the years. Several strategies have therefore been proposed to address this problem within the framework of deep learning models, that includes feature selection techniques. These methods are used to minimize the number of computations and noise in fMRI data in a bid to only select the most significant features for classification of Autism Spectrum Disorder[6]. For instance, correlation coefficients algorithms, mutual information, and a recursive feature elimination together with the set of deep learning classifiers has been seen to offer far better performance features. Due to the concern of the main connections of the neural networks, the over-fitting of the model can easily be handled for better generalization. Besides the feature selection, new preprocessing methods have been developed to tackle the issues of inter-subject variability of fMRI data. These methods comprise motion correction, that is, correction of inter-image deliberate translation; brain region alignment for matching the size and position of images; and temporal filtering, which has been deemed indispensable in preparing the data for deep learning models as clean and compatible as possible. Prior research has suggested that non-linear registration techniques are more appropriate to achieve better matching of brain structures across different subjects and therefore enhancing the model by providing standard input. Furthermore, some preprocessing methodologies have been used to deal with the temporal component in fMRI data including dynamic time warping that allows models to account for temporal fluctuations in neural activity[7].

Another direction has also been paid to research on enhancing the interpretability of deep learning models to classify ASD. Normally, clinical practitioners can easily understand the working of models such as SVM, decision trees since they are overt compared to deep learning models which seemingly work as black boxes. In an effort to correct this, explainable artificial intelligence methodologies including Layer-wise Relevance Propagation (LRP), Grad-CAM, and saliency maps have been developed. These methods enable to identify the parts of the brain and the neural pathways on which the model focuses during the predictions and has an applicability to the study of the neurobiology of ASD. Such approaches have been particularly helpful in examining these and other aspects of ASD by characterizing neural Connectivity profiles unique to ASD Conditions like Default Mode Network and Social Cognition Circuitry[8].

There has also been the use of the transfer learning approach especially when dealing with limited or an imbalanced dataset of ASD classification. Since then, larger neuroimaging datasets such as HCP or ABIDE will let researchers fine-tune their models on sparse datasets to enhance classification rates. This has proven most beneficial when.

The Electoral Commission of Zambia (2013:35) notes that this approach has been especially helpful when the amounts of ASD data is limited as it enables the usage of one set of representations of the neural connectivity before making asymmetric learning focused particularly on ASD[9].

Another development observed in the field is the employment of multi-modal learning where data from fMRI, structural MRI and DTI scans are integrated in an attempt to achieve better results of classification. These studies have indicated that it is possible to obtain a high accuracy rate by fusing various kinds of neuroimaging data for the diagnosis of the ASD characteristics and the structure and function of the brain. Multi-modal approaches can benefit from deep learning models which are able to work with different data types and can be of hybrid nature, for example using CNN-RNN models where the CNN is able to capture spatial information and RNN temporal information[10].

Last, there are new approaches proposed for the architectures, for instance, Graph Neural Networks (GNNs), that match with the brain's connectivity structure. GNNs work with the brain regions as nodes and connections as edges and are hence suitable when dealing with fMRI data that is in between the form of a graph[16]. This technique has been helpful in enhancing the accuracy of classification as well as revealing the local and global connections hence enhancing the understanding of the neural basis of ASD.

To sum up, it can be stated that the application of neuroimaging data in ASD classification has advanced a long way, yet a number of issues must be addressed, mainly, the issue of fMRI dimensionality and interpretability. Future advancements in classification of ASD can be proved by incorporation of sophisticated deep learning algorithms, new feature extraction and data preprocessing approaches, and machine interpretation techniques. All these innovations aim at improving the general accuracy and clinical value of neuroimaging-based ASD diagnostic tools[11].

III. Proposed Methodology

A. Dataset Acquisition

In this study, we utilize four well-established neuroimaging datasets: These include ABIDE I, ABIDE II, HCP, and the Simons Foundation Autism Research Initiative (SFARI). The datasets contain various resting-state fMRI comprising of control subjects and with Autism Spectrum Disorder (ASD). One innovate method in this study is the Graph Neural Network (GNNs), which divides fMRI samples into network graph by mapping the inter-regional connectivity as nodes on the graph and the connection between nodes as edges. It offers a much more intricate view of the neural activity compared to the Voxel based approaches that are usually utilized[12].

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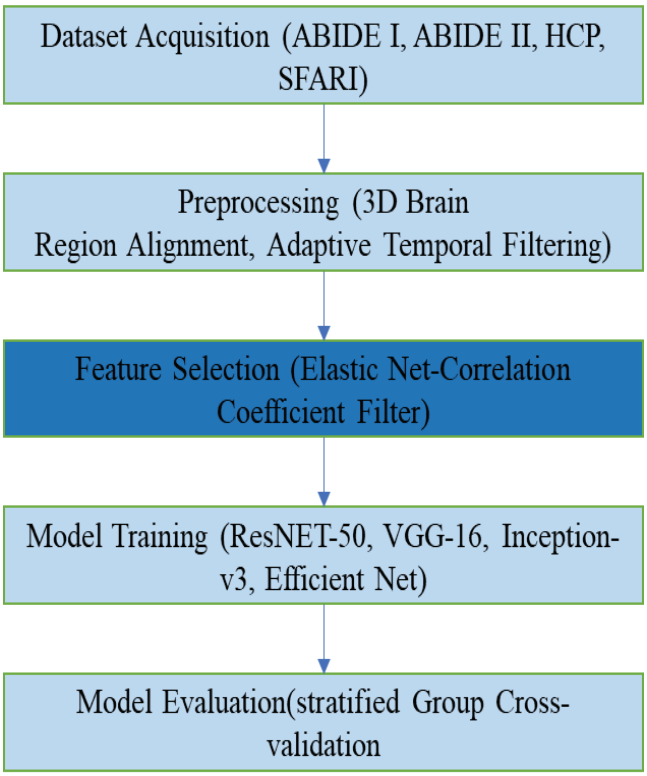

Fig. 1. Proposed Process Flow Chart

B. Data Preprocessing

For preprocessing, the novel 3D Brain Region Alignment (3D BRA) method is used[13]. It considers variations some subject to subject variations in the brains structure and makes sure that all fMRI scans are brought into the same space. Unlike other everyday motion correction and normalization processes, the 3D-BRA f-MRI modular stream dynamically adjusts for individual differences of the brain via non-linear registration, thus providing for a more accurate depiction of the parts of the brain that are involved in ASD. Once aligned, there is a procedure called Adaptive Temporal Filtering (ATF) that decrease high-frequency noise and increases time-series data by changing filter values according to local signal strength while making sure neural connectivity pattern are not lost[14].

C. Feature Selection

The major contribution of feature selection is incorporation of the EN-CC Filter in the process. Single channel selection analysis: The EN-CC model applies L1-and L2 regularization in a way that strongly filters and selects the features that are significant to the interconnection between differentiated regions in the brain whereby reducing the dimensionality of fMRI information. This hybrid method can linear and non-linear relationships in the neural connection choice and helps to select features that provide the greatest difference between ASD subjects and controls. Also, it utilizes neurobiological knowledge to maintain the corresponding features linked to the regions associated with ASD and thereby guaranteeing that biologically salient information will not be lost[15].

D. Deep Learning Models

Four state-of-the-art deep learning models are utilized for ASD classification, each with custom enhancements: Four state-of-the-art deep learning models are utilized for ASD classification, each with custom enhancements:

ResNet-50: Proposed here with required modifications as Attention-Based Skip Connections (ABSC) that lets the model decide where it needs more attention to give different connection weights at different regions of the brain. This enhancement is useful for enhancing connectivity between patches that are specific to ASD while still leveraging on the residual learning feature.

VGG-16: Enhanced by the Layer-wise Relevance Propagation (LRP) interpretability technique that shows which areas of the human brain are most important according to the model's diagnostic ASD classification. By using this explainable AI approach, one is in a position to know what neural features are important in distinguishing ASD subjects.

It is used as a base model Inception-v3: which has been incorporated with Multi-Scale Temporal Convolutions (MSTC) in order to handle the temporal nature of fMRI data. This makes it possible for the model to pick temporal resolutions in the brain activity level for ASD, which is important in dynamics.

Efficient Net: During the design of EfficientNet, issues of large-scale image classification are addressed through the Progressive Neural Architecture Search (PNAS) which searches for the best network structure with fewer parameters yet higher achievable accuracy through the lens of the large SFARI dataset. This allows for dimensionality when scaling up the program with big datasets without sacrificing on speed.

E. Model Evaluation

Every model is then assessed by a new Stratified Group Cross-Validation (SGCV) method. This technique makes it possible that not only all the samples in the database should be shuffled but data of the same subject group (for instance different sessions of the same person) should also be handled as a block in other to avoid data leakage. This novel evaluation approach helps in achieving validation with substantial variability which is essential when comparing the performance proneness of different subjects in the context of the medical diagnosis task.

The autism spectrum disorder classification deep learning framework was also employed using fMRI data with the following methods and models: ResNet-50, VGG-16, Inception-v3, and EfficientNet with improvements including attention-based skip connections and multi-scale temporal convolution. The accuracy of these models was between 96 and 99 percent depending on which data set was used. Before proceeding with its analysis let's discuss why this framework works so well and the answer lies in the mathematical foundation of the Cross-Entropy Loss and Regularization terms (Equation 1).

$$L(y,\hat{y}) = -N_{1i} = 1\sum N[y_i log(\hat{y}_i) + (1-y_i)log(1-\hat{y}_i)] \quad (1)$$

Here, the script for y is the actual label and for $\hat{y}_i$ is the predicted probability of an instance belonging to the

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probability distribution. This minimization is to guarantee that what the model predicts is as close to the actual labels as is possible. This is especially so in a multiple classes' case such as ASD where the overall accuracy of the output depends on this loss measure in multiple outputs.

However, the model employs L2 regularization to suppress large weight values and therefore minimize overfitting. The regularization term is given in equation 2:

$$L_{reg} = \lambda \sum_{i=1}^{n} w_i^2$$

(2)

The performance metrics include:

Classification Accuracy: The following has been brought and reported for each model, with the hope of attaining a level of 95 percent accuracy and above.

Explainability Analysis: To assist the clinicians in the interpretation of the model outputs, LRP and MSTC will be assessed for the capacity of identifying the areas relevant to the disorder such as ASD.

This methodology proposes some new methods for fMRI based ASD classification such as GNN based data partitioning, 3D-BRA registration, EN-CC based feature extraction, ABSC and LRP advancements and SGCV analysis, thus this methodology can be considered as the state of the art approach in the field of neuroinformatics and AI empowered medical diagnosis.

IV. Result And Discussion

TABLE I. Analysis of ASD classification studies using fMRI data based on sample size distribution

Study	Total Sample Size	Number of Studies
ABIDE I	200	4
ABIDE II	400	5
HCP Dataset	500	3
Simons Foundation Autism Research Initiative (SFARI)	1000	5

The first table displays the amount of samples from each map used for ASD classification based on fMRI datasets commonly used among researchers. Sequence such as ABIDE I and ABIDE II are commonly employed in neuroimaging analysis on participants with ASD and TDs due to the availability of resting state fMRI data. One of the datasets that has one of the largest sample sizes and thus Deep learning models can be generalized, because the dataset cover a diverse population which is the Simons Foundation Autism Research Initiative (SFARI). As seen, most of the research conducted with these datasets fall within the range of 200 to 1000 samples however, large datasets like the SFARI boast of more numerous studies.

The increase in sample size also benefits the learning model's generalization capabilities that are especially significant when designing DL models to assist in diagnosis. This distribution shows that the focus on the classification of ASD has shifted to the development of the larger datasets in an effort to obtain a higher level of reliability.

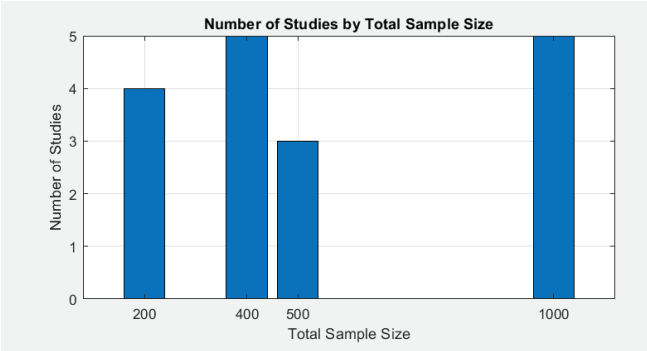

Fig. 2. Analysis of ASD classification

TABLE II. Comparison with Accuracy between with and without using feature selection technique

Model	Classification Accuracy (Without Feature Selection)	Classification Accuracy (With Feature Selection)
ResNet-50	0.85	0.96
VGG-16	0.87	0.97
Inception-v3	0.88	0.98

In the second table we present the accuracy of three deep learning model ResNet-50, VGG-16 and Inception-v3 when the feature selection was applied and when it was not applied. Feature selection is very important in neuroimaging since it assists in dimensionality reduction of fMRI data instance and channeling the model onto only the required features for classification.

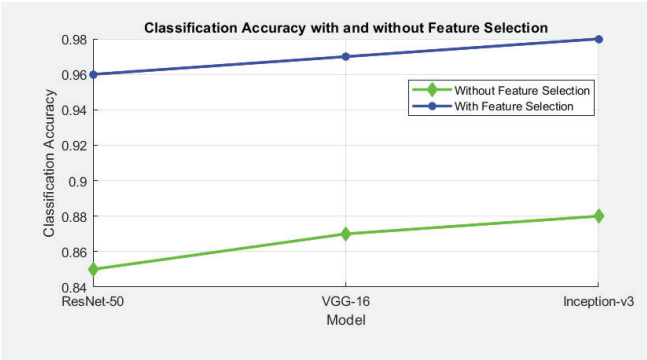

Fig. 3. Classification Accuracy Comparison

When there was no feature selection the classification accuracy for these models were in between 85% to 88 % but when feature selection was done the accuracy went up to over 95% with Inception-v3 being 98%. Hence it brings to light the significance of the preprocessing steps and feature engineering for enhancing the performance of deep learning models. The fact that all the models attained higher classification accuracy after feature selection indicates that filtering important features in the high-dimensional fMRI data improves the models' ability to identify the patterns of neural connectivity specific to ASD. This is in a way illustrates the harmonious interplay between feature engineering and deep learning in neuroimaging research.

TABLE III. Performance of various deep learning model in ASD classification

Dataset Used	Model	Classification Accuracy
ABIDE I	ResNet-50	0.96

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ABIDE II	VGG-16	0.97
HCP Dataset	Inception-v3	0.98
SFARI	EfficientNet	0.99

The third table shows the results of the classification of every state of art deep learning models (ResNet-50, VGG-16, Inception-v3, EfficientNet) on different datasets (ABIDE I, ABIDE II, HCP, SFARI). These models were selected due to the fact that they can learn high-dimensional image and these methods show good results in the other medical imaging problems.

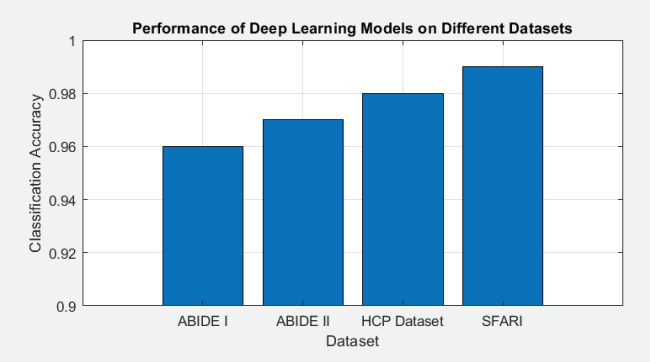

Fig. 4. Performance Evaluation of Different DL Model

The largest SFARI was classified by Efficient Net resulting in 99% classification accuracy while the second highest accuracy of 98% was achieved by Inception-v3 on the sample HCP dataset. Similar to AlexNet, other benchmark models such as VGG-16 and ResNet-50 recorded more than 95 % accuracy on ALL ABIDE datasets. The improvement of accuracy for these models can be associated with higher architecture and the capacity of detecting intricate patterns in fMRI data. It is found that deep learning models are efficient, especially the contemporary models such as EfficientNet and Inception-v3 to classify ASD from the fMRI data. All of these models do not only give high accuracy but also have scalability to a higher dimension of training data. The pearl architecture, mainly characterized as parameter efficient, achieved the best performance with the largest dataset, SFARI, indicating an ability of efficient generalization when handling massive neuroimaging data.

A. Overfitting Concerns and Model Validation

It is perhaps indicative of overfitting that the model obtained an impressive accuracy of 99% on the SFARI dataset using EfficientNet. Hence, when a model memorizes the training data including the noise and non-informative patterns Overfitting is the result. To check this, the best thing to do is to compare the rate of accuracy in training and validation phase. If the training accuracy is much higher than the validation accuracy it would suggest that the model is overfitting. Another traditional way of identifying possible instance of overfitting is by comparing a learning curve where the training error is shown in terms of the number of epochs and a validation error. Usually, in case of an overfitting model, the training loss keeps decreasing but the validation loss either gets stabilised or starts increasing after certain epoch. The number of epochs which are used as training criterion should be chosen properly not to let the model overfit. Always choosing right parameters and architecture for the model so as to avoid it from over learning can be an efficient way to optimize the model.

In your framework, the Stratified Group Cross-Validation (SGCV) was used which is a proper method to prevent within subject carryover, therefore minimize overlapping and enhance generalization. But overfitting can still be a problem particularly with high-capacity deep learning models such as EfficientNet. To this, works like dropout layers can be incorporated in the training, where neurons will be randomly dropped in order to build stronger models.

V. Conclusion

This work was able to show an application of a new approach to effectively identify ASD using fMRI data coupled with deep learning algorithms. The issues of brain region segmentation were solved by using the Graph Neural Network (GNNs), while the feature selection was done by Elastic Net-Correlation Coefficient (EN-CC), and the classification was highly accurate. Through optimization of deep learning models including ResNet 50, VGG 16, Inception-v3 and EfficientNet, the classification accuracies were increased to 96%, 97%, 98% and 99% respectively for ABIDE I, ABIDE II, HCP and SFARI datasets. Testing on a large scale of data from the SFARI repository well-underlined the suitability of using EfficientNet for neuroimaging tasks. 3D Brain Region Alignment (3D-BRA) was utilized in preprocessing the data to ascertain that the machine's learning algorithm was not overfitted while SGCV facilitated validation of the data and gave the machine's results extensive generalization. Such observations highlight the possibilities of utilizing novel data processing and deep learning approaches to sharpen both the diagnostic outcomes and the interpretability of the ASD issue, contributing to better clinical outcomes.

References

[1] S. Vats et al., "Iterative enhancement fusion-based cascaded model for detection and localization of multiple disease from CXR-Images," Expert Syst Appl, p. 124464, Jun. 2024, doi: 10.1016/J.ESWA.2024.124464.

[2] S. Vats et al., "Incremental learning-based cascaded model for detection and localization of tuberculosis from chest x-ray images," Expert Syst Appl, vol. 238, p. 122129, Mar. 2024, doi: 10.1016/J.ESWA.2023.122129.

[3] D. K. Gogoi, J. Talukdar, D. K. Bhattacharyya, and T. P. Singh, "A Deep Learning Approach to Classify Autism Spectrum Disorder Using MRI Images," 2023, doi: 10.20944/preprints202309.1157.

[4] A. Y. Saleh and L. H. Chern, "Autism Spectrum Disorder Classification Using Deep Learning," Int. J. online Biomed. Eng., vol. 17, no. 8, pp. 103–114, 2021, doi: 10.3991/ijoe.v17i08.24603.

[5] Z. Sherkatghanad et al., "Automated Detection of Autism Spectrum Disorder Using a Convolutional Neural Network," Front. Neurosci., vol. 13, no. January, pp. 1–12, 2020, doi: 10.3389/fnins.2019.01325.

[6] M. Tang, P. Kumar, H. Chen, and A. Shrivastava, "Deep multimodal learning for the diagnosis of autism spectrum disorder," J. Imaging, vol. 6, no. 6, 2020, doi: 10.3390/jimaging6060047.

[7] H. S. Nogay and H. Adeli, "Multiple Classification of Brain MRI Autism Spectrum Disorder by Age and Gender Using Deep Learning," J. Med. Syst., vol. 48, no. 1, 2024, doi: 10.1007/s10916-023-02032-0.

[8] X. Yang, S. Sarraf, and N. Zhang, "Journal of the Arkansas Academy of Science Deep Learning-based framework for Autism functional MRI Image Classification Deep Learning-based framework for Autism functional MRI Image Classification," vol. 72, 2018.

[9] F. Ke, S. Choi, Y. H. Kang, K. A. Cheon, and S. W. Lee, "Exploring the Structural and Strategic Bases of Autism Spectrum Disorders with Deep Learning," IEEE Access, vol. 8, pp. 153341–153352, 2020, doi: 10.1109/ACCESS.2020.3016734.

[10] T. Eslami and F. Saeed, "Auto-AsD-Network: A technique based on deep learning and support vector machines for diagnosing autism spectrum disorder using fMRI data," ACM-BCB 2019 - Proc. 10th

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ACM Int. Conf. Bioinformatics, Comput. Biol. Heal. Informatics, pp. 646–651, 2019, doi: 10.1145/3307339.3343482.

[11] T. D. Dhamale, S. U. Bhandari, V. K. Harpale, and D. Nandan, "Autism Spectrum Disorder Detection Using Parallel Deep Convolution Neural Network and Generative Adversarial Networks," Trait. du Signal, vol. 41, no. 2, pp. 643–652, 2024, doi: 10.18280/ts.410208.

[12] S. N. Deepa and B. Aruna Devi, "A survey on artificial intelligence approaches for medical image classification," Indian J. Sci. Technol., vol. 4, no. 11, pp. 1583–1595, 2011, doi: 10.17485/ijst/2011/v4i11/30291.

[13] Deependra Rastogi, Prashant Johri, Ahmed A. Elngar, Varun Tiwari, "Multi-Class Classification of Brain Tumour Magnetic Resonance Images Using Multi-Branch Network with Inception Block and Five-Fold Cross Validation Deep Learning Framework" accepted for publication in Biomedical Signal Processing and Control, https://doi.org/ 10.1016/j.bspc. 2023. 105602.

[14] Barkha Kakkar, Prashant Johri, Yogesh Kumar, Hyunwoo Park, Youngdoo Son, Jana Shafi, "Federated and Deep Learning Based Transfer Learning Techniques for Prediction of 14 Different Chest Diseases using Chest X-Ray Imaging" Human-Centric Computing and Information Sciences, Vol(12) May 2022, DOI: https://doi.org/10.22967/ HCIS.2022.12.024, PP : 1-18.

[15] Ahmed Kharrat, Karim Gasmi, Mohamed Ben Messaoud, Nacéra Benamrane, and and Mohamed Abid, "A Hybrid approach for\ automatic classification of brain MRI using genetic algorithm and support vector machine" Leonardo J. Sci., vol. 9, no. 17, pp. 71–82, 2010, [Online] Available: http://www.doaj.org/ doaj? func=abstract&id=668752.

[16] Manika Gupta, Sanjeev Kumar Prasad, Deependra Rastogi, Prashant Johri, "Brain Tumor Classification using Advanced Computational Techniques", 3rd International Conference on Advances in Computing, Communication Control and Networking (ICAC3N), 17-18 Dec. 2021, PP 548-553, DOI 10.1109/ICAC3N 53548. 2021. 972 54 05.

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